

Lab in the Cloud for IoN-DRI @ UCL

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Scientific Computing

Challenge

- Research labs' need for computing resources (hardware, software, users) changes according to grants and recruitment
- Researchers requires independent and decentralised solutions
- Research excellence requires reproducible solutions

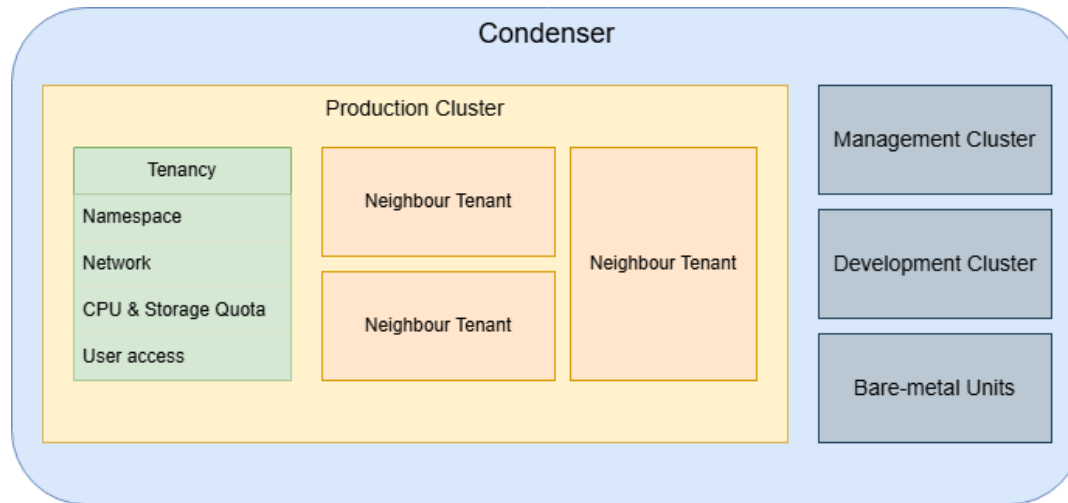
**How can we provide
dynamic and heterogeneous services
efficiently and sustainably?**



PaaS for Scientific Computing

Cloud-native approach

- Enable us to dynamically provision computing resources → **scalability and sustainability**
- [Cloud @ UCL – Condenser](#)



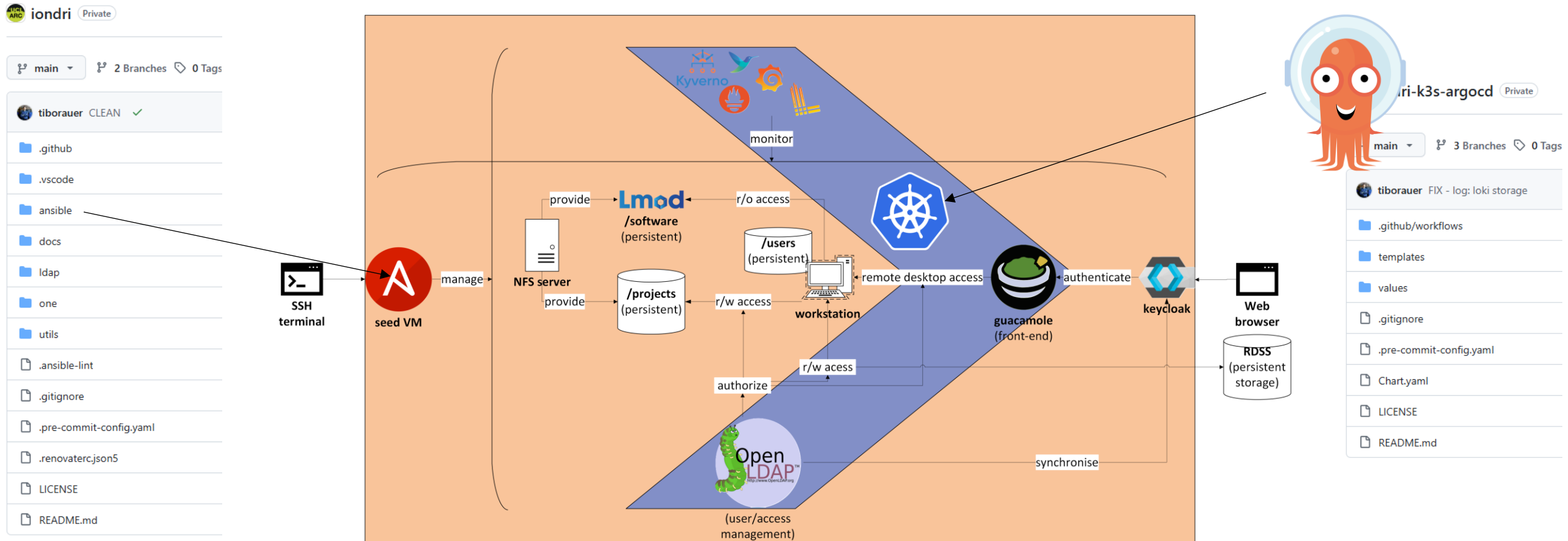
GitOps

- Enable us to use code as a single source of truth for infrastructure → **automatization and reproducibility**



Lab in the Cloud

Implementation for IoN-DRI @ UCL



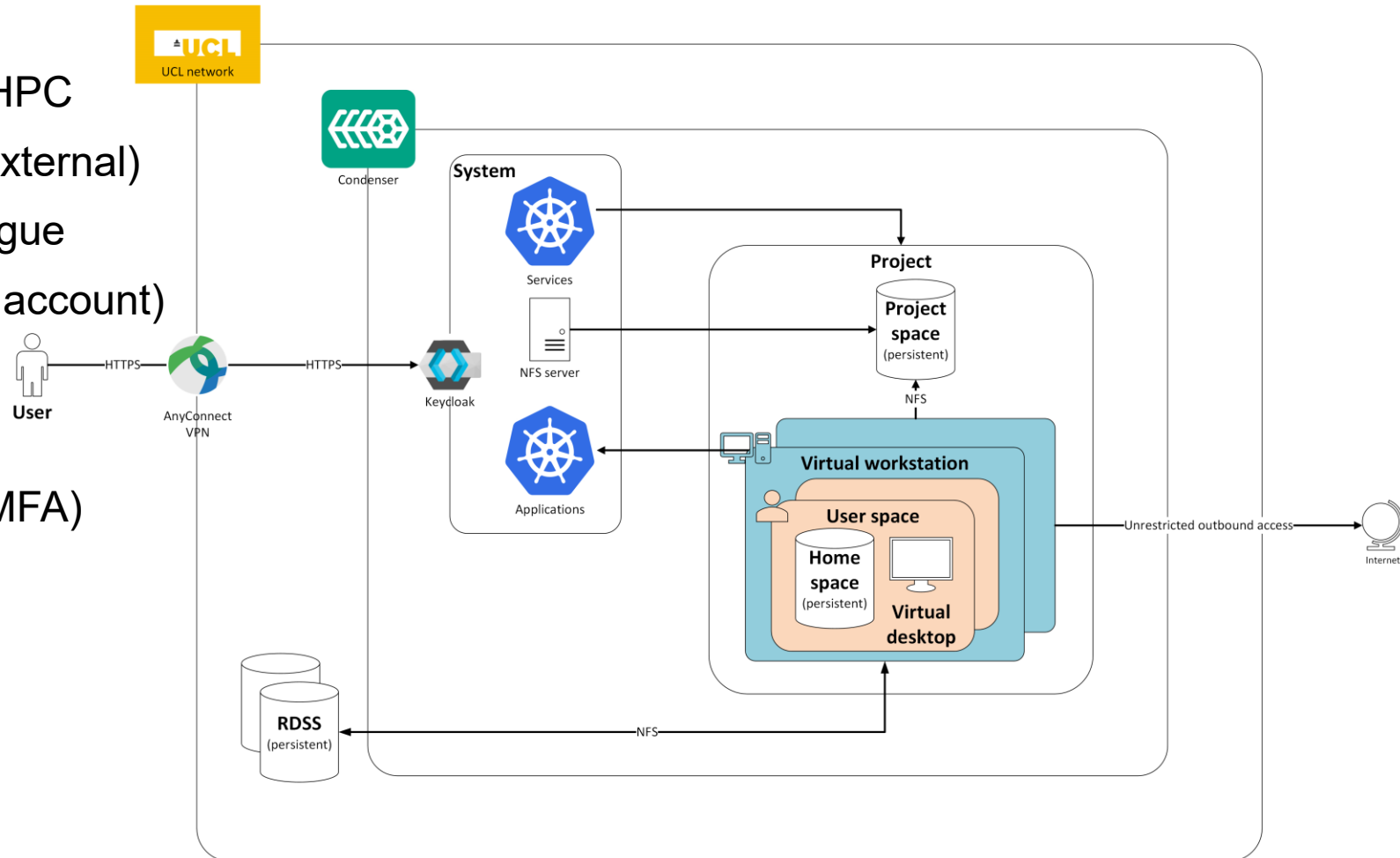
Lab in the Cloud

Current state

- Versatile resources: workstations, K8s, HPC
- Access to storage (within-platform and external)
- Pre-configured research software catalogue
- Accessible from web browser (with UCL account)

Future

- Security hardening (e.g., krb5s, mTLS, MFA)
- ISMS coverage for sensitive data (CRE)
- Automated and scalable data pipelines



Lab in the Cloud

Front-ends

- Documentation on SharePoint:
[*IoN-DRI Computing Platform*](#)
- User access via web browser:
[*Workstations portal*](#)
[*Self-service password portal*](#)
- Maintenance
[*LDAP Admin portal*](#)
[*Monitoring portal*](#)
[*Alerts on Slack*](#)



Overview

We are developing the IoN-DRI Computing Platform, which will be available for IoN-DRI researchers, as part of the Scientific Platform which is being developed for the IoN-DRI programme.

The IoN-DRI Computing Platform provides computing resources tailored to various specific research needs and ensures the robustness and scalability of these computing resources. These resources can include workstations with desktop, application servers, as well as, clusters for high-performance computing, Kubernetes, and more.

Maintainers



Tibor Auer
Senior Research Infrastructure Developer (Faculty of Brain...

Resources

Workstations

- Independent machines with dedicated resources (CPU, RAM, disk)
- Desktop for interactive work

Software

- Docker/Apptainer
- Python (miniforge)
- VS Code
- MATLAB
- OCTAVE
- FreeSurfer
- FSL
- dcm2niix
- DCMTK

Storage

- User space (machine-specific)
- Collaborative project space shared across the machines
- Mounted RDSS

Access

Please note, that access is only available via the UCL campus network or through the UCL VPN.

Please, contact one of the [maintainers](#) to register your UCL username and provide with an (initial) password for the platform. Then, you can use these credentials or **your UCL account** to access the links below.

Platform

